MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, POWER

TYPE 2N158

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a PNP, germanium, power transistor.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

P_T 1/	V_{CBO}	V_{EBO}	V_{CES}	I_E	T_A
W	*Vdc*	*Vdc*	*Vdc*	*Adc*	*°C*
17	-60	-30	-60	2	-65 to +100

1/ Derate linearly 0.283 W/°C for $T_A > 25°$ C.

1.4 Primary electrical characteristics.

Limits	f_{hfe} V_{CE} = -2 Vdc I_C = -0.5 Adc	h_{FE} V_{CE} = -2 Vdc I_C = -0.5 Adc	h_{FE} V_{CE} = -2 Vdc I_C = -1.0 Adc	$V_{CE(sat)}$ I_B = -150 mAdc I_C = -1.0 Adc
	kHz			*Vdc*
Min	4	21	13	---
Max	---	---	40	-0.75

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - *Semiconductor Devices, General Specification for.*

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STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of group A, B, and C inspections.

4.3.1 *Group A inspection.* Group A inspection shall consist of the examinations and tests specified in table I.

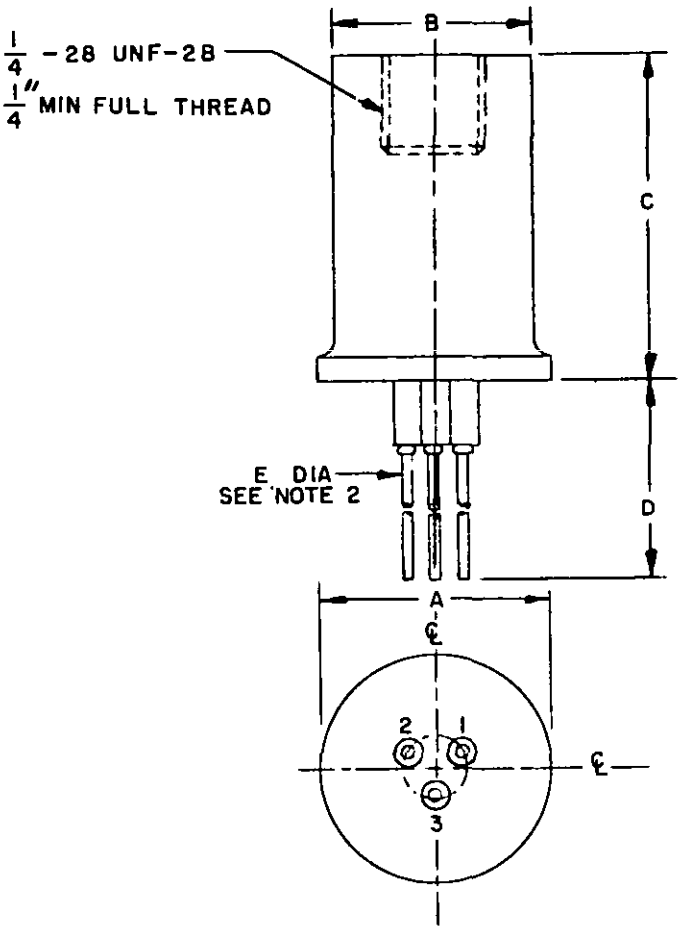
4.3.2 *Group B inspection.* Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 *Group C inspection.* Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

4.3.4 *Group B and group C life-test samples.* Samples that have been subjected to group B, 340-hour life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria (see 4.3.3).

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III.

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LTR	DIMENSIONS			
	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.530	.580	13.46	14.73
B	.415	.525	10.54	13.34
C	.690	.828	17.53	21.03
D	2.000	---	50.80	---
E	.021	.025	.53	.64

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Three leads equally spaced.
3. The lead arrangement shall be as follows:
   Lead 1.....Collector.....Black
   Lead 2.....Base.....White
   Lead 3.....Emitter.....Red
4. The collector shall be electrically connected to the case.

FIGURE 1. Physical dimensions of transistor type 2N158.

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TABLE I. *Group A inspection*

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			—	—	—	—
Subgroup 2			5				
Breakdown voltage, collector to base	3001	Bias cond D; I_C = -1.0 mAdc		BV_{CBO}	-60	—	Vdc
Breakdown voltage, collector to emitter	3011	Bias cond. D; I_C = -250 mAdc		BV_{CEO}	-30	—	Vdc
Collector to emitter cutoff current	3041	Bias cond C; V_{CE} = -60 Vdc		I_{CES}	—	-1.5	mAdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -2 Vdc		I_{CBO}	—	-100	μAdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -60 Vdc		I_{CBO}	—	-1.0	mAdc
Emitter to base cutoif current	3061	Bias cond. D; V_{EB} = -30 Vdc		I_{EBO}	—	-0.5	mAdc
Subgroup 3			5				
Base emitter voltage (nonsaturated)	3066	Test cond B; V_{CE} = -2 Vdc; I_B = -0.5 Adc		V_{BE}	—	0.85	Vdc
Base emitter voltage (saturated)	3066	Test cond A; I_B = -150 mAdc; I_C = -1.0 Adc		$V_{BE(sat)}$	—	-1.3	Vdc
Collector to emitter voltage (saturated)	3071	I_B = -150 mAdc; I_C = -1.0 Adc		$V_{CE(sat)}$	—	-0.75	Vdc
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -0.5 Adc		h_{FE}	21	—	—
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -1.0 Adc		h_{FE}	13	40	—
Subgroup 4			10				
Small-signal short-circuit forward-current transfer-ratio cutoff frequency	3301	V_{CE} = -2 Vdc; I_C = -0.5 Adc		f_{hfe}	4	—	kHz

TABLE II. *Group B inspection.*

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Physical dimensions	2066	(See figure 1)		—	—	—	—
Subgroup 2			15				
Solderability	2026			—	—	—	—
Thermal shock (temperature cycling)	1051	Test cond. B, except T_{max} = +100°C		—	—	—	—

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TABLE II. *Group B inspection - Continued*

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2 - Continued							
Thermal shock (glass strain)	1056	Test cond. A		---	---	---	---
Seal (leak rate)	1071	Test cond. G or H for fine leaks; test cond. A, C, D or F for gross leaks		---	---	$5x10^{-7}$	atm cc/sec
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -60 Vdc		I_{CBO}	---	-1.0	mAdc
Emitter to base cutoff current	3061	Bias cond. D; V_{EB} = -30 Vdc		I_{EBO}	---	-0.5	mAdc
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -1.0 Adc		h_{FE}	13	40	---
Subgroup 3			15				
Shock	2016	Nonoperating; 500 G, 1 msec, 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration, variable frequency	2056	100 to 1000 Hz; 10 G		---	---	---	---
Constant acceleration	2006	5000 G in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test cond. E		---	---	---	---
End points:							
Seal (leak-rate)	1071	Test cond. G or H for fine leaks; test cond. A, C, D or F for gross leaks		---	---	$5x10^{-7}$	atm cc/sec
Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 6			7				
High-temperature life (nonoperating)	1032	T_{stg} = +100°C; time = 340 hours (see 4.3.4)		---	---	---	---

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TABLE II. *Group B inspection - Continued*

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 6 - Continued							
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -60 Vdc		I_{CBO}	---	-2.0	mAdc
Emitter to base cutoff current	3061	Bias cond. D; V_{EB} = -30 Vdc		I_{EBO}	---	-1.0	mAdc
Forward-current transfer ratio	3076	V_{CE} = -2 Vdc; I_C = -1.0 Adc		h_{FE}	8	50	---
Subgroup 7			10				
Steady-state operation life	1027	T_C = +60° ±5°C; V_{CB} = -14 Vdc; P_T = 7 W; time = 340 hours (see 4.3.4)		---	---	---	---
End points:							
(Same as subgroup 6)							

TABLE III. *Group C inspection*

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Thermal resistance	3131			θ_{J-C}	—	3.5	°C/W
Subgroup 2			λ=10				
High-temperature life (nonoperating)	1031	T_{stg} = +100°C (see 4.3.4)		---	—	---	---
End points:							
(Same as subgroup 6 of group B)							
Subgroup 3			λ=15				
Steady-state operation life	1026	T_C = +60° ±5°C; V_{CB} = -14 Vdc; P_T = 7 W (see 4.3.4)		—	---	---	---
End points:							
(Same as subgroup 6 of group B)							

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5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES:

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU, MI
Navy - SH
Air Force - 11, 17
DSA - ES

User activities:
Army - SM
Navy - CG, MC, OS, AS
Air Force - 13, 15, 19, 70, 80

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5961-0156)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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